Index-calculus baselines: the ratchet

Status (2026-10-15). This is the first edition of the ledger, with the tuned-rho rows added from run 38777903922. It takes no new measurement: every number is copied from a committed experiment, and each row cites that experiment. Figure 1 was checked and stays current: no baseline row's numbers changed. The n=83 rows for lift_scan, the Feistel order and their (not promoted) combination were added on 2026-10-15; Figure 1 was checked and stays current, because no baseline row's numbers changed. The n=83 row for the probe pipeline was added on 2026-10-16, the two-level root table and the state fold the same day, each with the same Figure 1 check. On 2026-10-16 the stacked producer (`koblitz-n83-stacked-20261016`, run 38862448935) replaced B's n=83 baseline row, and Figure 1 was re-rendered. The machine-readable twin is `baselines.json`; update both files in the same change.

The rule this file serves

AGENTS.md, "Success criterion: iterative improvement of the index-calculus baseline" (owner, 2026-10-15; cryptanalysis#919, crypto#1908):

- **The baseline** for a curve and size is the best current verified IC pipeline. It is frozen with its protocol, receipts and end-to-end cold `S`.
- **A success** is any verified, measured improvement over that baseline on the same instances, under a protocol frozen before timing. Advance and engineering both count. Relabelling and accounting are recorded and do not count.
- **The ratchet.** A landed improvement replaces the row here, and the old row moves to the history with its delta.
- **The gap** to the matched rho, and to Bernstein–Lange (BL) where it was measured, is the distance remaining. It is not a pass/fail gate. A claim to *beat* rho still needs every reference rule: a cold `S` below the matched rho's, robust at growing `n`, independently replayed.

How to read the numbers.

- **One row, one experiment.** No row combines numbers from two experiments. Where an experiment did not measure a quantity, the cell says so.
- **Every gap is IC cost over the reference's cost.** Above 1 is distance still to close; below 1 means the IC is cheaper on that boundary. The sources quote the online figure the other way up (rho over IC). That figure is kept in each JSON row's `source_figure`.
- **`S` = cost / $\sqrt{\mathbf{r}}$**, in the row's own unit. The unit differs between experiments, so compare `S` inside a row only. The gaps carry no unit.
- **Tuned rho** means rT770: A's tuned `StrongRhoG` (Fold field kernel, 64 lanes, DP 6, escape off). `koblitz-n83-reconcile-20261014` found it to be the fastest correct n=83 reference. At n=41–61 it has been run as rT770's configuration on the one-word kernel (`koblitz-s23-tuned-rho-20261015`). ParallelRho, inside crypto's `ic price`, is crypto's own signed-Frobenius walk (64 lanes in lockstep). It is the matched rho at n=41–61, but it is not rT770.

The baselines

n	curve	current best verified IC pipeline (commit; protocol)	targets	`S` cold / online (IC)	`S` cold / online (rho)	cold gap, IC / rho [95%]	online gap, IC / rho [95%]	gap to tuned rho (rT770)	gap to BL (online, IC / BL)	evidence tier	next improvement to try
19	E0 / GF(2^19), a=1	volcano-ic τ-orbit dim-6 base, Sage Gröbner (arm `tau6`); cryptanalysis `3f722e2e`; protocol.json of volcano-ic/ rho-arm-20261014; run 38520305068	10 planted scalars, each one complete ECDLP	5,154 / not separable	0.260 / —	**1.94e4** [1.69e4, 2.23e4] (CPU, geometric mean)	not measured: each IC run is cold	none; the matched rho is a signed-Frobenius walk at 0.997× the floor in steps, run in Python/Sage, not tuned for wall time	none	exploratory (GitHub VM, unpinned)	Move tau6's per-relation Gröbner solve out of Sage (native, or the subspace `PDP2ht` oracle); re-run against the same rho arm

n	curve	current best verified IC pipeline (commit; protocol)	targets	S cold / online (IC)	S cold / online (rho)	cold gap, IC / rho [95%]	online gap, IC / rho [95%]	gap to tuned rho (rT770)	gap to BL (online, IC / BL)	evidence tier	next improvement to try
41	K_0 / GF(2^41)	`ic price --single-target` at IC programme baseline v3: 6,560 points, 80 columns, aimed m=3 collection, m=2 pair-table descent; crypto `995ea207`; rule v3	64 public points, 1 process each, +4 A/A	11.17 / 0.121	1.459 / 1.318	**7.66** [6.90, 8.61]	0.092 [0.071, 0.116]	**measured in a separate experiment, not combined**: see the 41 row of Other verified records (run 38777903922: online IC/tuned 0.09, cold 6.6); the matched rho in this row is ParallelRho (0.56× the strong walk's ns/step)	model only: 2.21	crypto isolated_bench container	Setup is 98.9% of cold S: cut the reusable setup at fixed recipe, as a new IC programme round against v3
53	K_0 / GF(2^53)	same pipeline, 27,984 points, 264 columns; crypto `995ea207`; rule v3	64 + 4 A/A	16.98 / 0.058	1.282 / 1.236	**13.2** [11.8, 15.0]	0.047 [0.036, 0.059]	**measured in a separate experiment, not combined**: see the 53 row below (online 0.0423, cold 10.6); ParallelRho here is 0.53× strong	model only: 1.77	crypto isolated_bench container	Setup is 99.7% of cold S: cut it at fixed recipe; then price the n=53 representative-plus-Frobenius orbit domain as its replacement
61	K_0 / GF(2^61)	same pipeline, 39,040 points, 320 columns; crypto `995ea207`; rule v3	64 + 4 A/A	24.93 / 0.079	1.189 / 1.168	**21.0** [18.6, 23.8]	0.067 [0.051, 0.088]	**measured in a separate experiment, not combined**: see the 61 row below (online 0.0687, cold 18.9); ParallelRho here is 0.50× strong	model only: 3.52	crypto isolated_bench container	Setup is 99.7% of cold S, and the cold gap grows as r^0.18: cut the setup; price the compact-orbit K=500 producer on these 64 points
83	K_1 / GF(2^83)	the stacked producer (arms cS / kS): B's `koblitz_orbit_dlp_fast_online` + `ic_log_cache.patch`, K600 (99,600 points, 600 orbit columns), lift_scan x Feistel + prefetched probes + 16-byte compact root slots + folded S3 root index (14,940,300 stored states), rank stage on the stack's own probe loop (candidate ID below); crypto `770f6a69`; `koblitz-n83-stacked-20261016/protocol.json` at cryptanalysis `24abca74`; run 38862448935	20 new public points, two rank blocks, Q0	9.680 / 0.01314 (rT770 steps, 316.66 ns)	0.1025 / 0.0944	**94.4** [73.5, 125.3] (end-to-end cold: rank stage + setup + online)	0.139 [0.077, 0.237]	**measured on this row**: end-to-end cold 94.4, online 0.139	not measured on this row (B's 235 [61, 645] is in Other verified records)	Tier 1 (GitHub runner, one pinned CPU)	The rank stage is 257.8 s of the 283.8 s end-to-end cold cost (91%): cut it (the guided minimum-rank policy, fewer attempts), then setup (lift_scan in the index build, 1.33x cold without rank); re-run the BL arm on the stack

Figure 1. The baseline rows' gaps on a log scale: cold and online to the matched rho, and online to Bernstein–Lange (a model; the one measured n=83 BL gap is in B's row, now under Other verified records). Each row is one experiment. Bars are 95% intervals where the source gives one (source, vector).

Curves and candidates.

n	curve ID	ICV1 slug	r	IC1 candidate of the baseline
19	`EC1N19Ckb1h01617b86df6c`	(volcano-ic surface curve E0)	130,873	none recorded
41	`EC1N41Ce0he09550ab560a`	`icv1-f2m41-tm2308219-7f48b14a`	549,756,390,943	`IC1N41Ckb0fb6560PDP3pairtableRCaimedLAsparseTDwalkISO0h5b25909970b3`
53	`EC1N53Ce0hb097de99be9a`	`icv1-f2m53-tm56619371-dac20a85`	21,044,858,204,113	`IC1N53Ckb0fb27984PDP3pairtableRCaimedLAsparseTDwalkISO0h715dd7692112`
61	`EC1N61Ce0hd8c15448081e`	`icv1-f2m61-t158598901-ab42b6c5`	162,888,033,982,417	`IC1N61Ckb0fb39040PDP3pairtableRCaimedLAsparseTDwalkISO0he9352435e473`

n	curve ID	ICV1 slug	r	IC1 candidate of the baseline
83	`EC1N83Ce1h35c7ced6b497`	`icv1-f2m83-t6151469093347-9eba6cd5`	8,569,786,107,849,059	`IC1N83Ckb1fb99600PDP4s3rootindexRCguidedLAdenseTDsingleISOOhc0ecc1fffbd7`

Notes on the rows.

- **n=19.** The `S` unit is one rho group operation of CPU time (2.17e-5 s). The IC's cold `S` is the τ6 arm's CPU per ECDLP in that unit. The baseline subspace base reads 1.77e5× rho in the same run. τ6 over the baseline base is 9.12 [8.12, 10.27].
- **n=41–61.** The unit is one batched affine addition, measured in each row's own process. v3's unit is 2.3–3.2× cheaper than §23's, so `S` is not compared across the two (rule v3 README). Every row passes `validate_claim` at `vs_rho` (68/68 per size). Both certificates were replayed outside the process, and the logarithms equal §23's.
- **n=83.** `S` is derived inside the one run: wall ms divided by that run's measured rT770 step (180.8 ns) and by $\sqrt{r}$. Cold is the matched precomputation P_IC (520,825 ms rank stage + 29,022 ms per-process setup) plus the mean online time. BL's precomputation was matched to P_IC in wall time on the same CPU, so the cold IC/BL ratio is 0.987 by construction. The baseline is this experiment, not the merged reconcile run, because it is the one experiment that measures the tuned rho, BL and cold together. If PR #918 changes, this row changes with it.

Other verified records

These rows are kept for context and as the history behind the baselines. None is combined with another row.

n	role	experiment	reference	cold gap, IC / ref [95%]	online gap, IC / ref [95%]	BL	tier
41	previous baseline (v0)	ledger §23, crypto `0bf67f16`	ParallelRho, same process	8.04 [7.26, 9.01]	0.107 [0.082, 0.135]	model 2.50	crypto isolated_bench container
53	previous baseline (v0)	ledger §23	ParallelRho	14.7 [13.1, 16.7]	0.057 [0.044, 0.073]	model 2.02	same
61	previous baseline (v0)	ledger §23	ParallelRho	31.6 [28.2, 35.7]	0.113 [0.084, 0.148]	model 8.04	same
41	other pipeline, complete cold CPU	crypto `disjoint_cold_v2_outcome_20261001` (run 36803331080), compact-orbit K85	32-walk signed-Frobenius normal-basis rho (not rT770)	13.4 [9.37, 18.5]	not measured	none	hosted, A/A and quiet-host gates, second-host replay
53	other pipeline, complete cold CPU	same, compact-orbit K220	same	22.7 [17.9, 31.6]	not measured	none	same
61	other pipeline	crypto autolab run `20261012T202048Z-5521627904`, compact-orbit K=500	strong rho rung 3 (not rT770)	83.4 (median of rows)	0.072 [0.052, 0.102] (inverse of median 13.94)	none	taskset-pinned container with a CPU-isolation receipt
41	gap to the tuned rho (this experiment only; cf. the baseline row)	`koblitz-s23-tuned-rho-20261015`, run 38777903922 (PR #923)	tuned = rT770 configuration on the one-word kernel; same run: ParallelRho cold 7.83, strong 6.43, rA770 2.54	**6.6** [5.91, 7.4]	**0.09** [0.0715, 0.111]	model 0.53	Tier 1
53	gap to the tuned rho (this experiment only; cf. the baseline row)	`koblitz-s23-tuned-rho-20261015`, run 38777903922 (PR #923)	tuned = rT770 configuration on the one-word kernel; same run: ParallelRho cold 11.8, strong 9.2, rA770 3.01	**10.6** [9.45, 12]	**0.0423** [0.0325, 0.0534]	model 0.322	Tier 1
61	gap to the tuned rho (this experiment only; cf. the baseline row)	`koblitz-s23-tuned-rho-20261015`, run 38777903922 (PR #923)	tuned = rT770 configuration on the one-word kernel; same run: ParallelRho cold 19.4, strong 13.7, rA770 4.42	**18.9** [16.7, 21.5]	**0.0687** [0.051, 0.0903]	model 0.759	Tier 1
83	previous baseline (B, the BL-arm row), superseded by the stacked producer	`koblitz-n83-bl-arm-20261014`, run 38689817618 (PR #918, not merged yet)	rT770, B's 16 fresh points + Q0 x4	360 (no interval; S 32.93 / 0.083 in that run's rT770 steps)	1.01 [0.30, 2.12], not separated	**measured: 235** [61, 645], 16/16 points BL faster; cold IC/BL 0.987 by construction	Tier 1
83	same pipeline as the baseline (merged)	`koblitz-n83-reconcile-20261014`, run 38617911080	rT770, two seeds per point	313 (fresh-rank setup)	0.92 [0.28, 1.78], not separated	none	Tier 1

n	role	experiment	reference	cold gap, IC / ref [95%]	online gap, IC / ref [95%]	BL	tier
83	same pipeline, first Tier 1 run	`koblitz-single-target-n83-tier1-20261014`, run 38523707244	rung 3, slower per step than StrongRhoG, so not the matched tuned rho	134	0.36 [0.11, 0.66]	model 33.5 (measured later: 235)	Tier 1
83	promoted single improvement over B: two-limb `lift_scan` canonicalizer (this experiment only)	`koblitz-n83-liftscan-20261015`, run 38776966036 (PR #926)	rT770, B's 16 fresh points	cold not measured	IC/rho 0.67 [0.19, 1.38] (inverse of 1.4914 [0.7270, 5.1445]); online full / lift_scan 1.5152 [1.5061, 1.5247]	none	Tier 1
83	promoted single improvement over B: Feistel scan order (this experiment only)	`koblitz-n83-scanorder-20261015`, run 38786433493 (PR #927)	rT770, B's 16 fresh points	cold not measured	IC/rho 0.291 [0.163, 0.481] (inverse of 3.4370 [2.0774, 6.1418]); online left-major / Feistel 3.4287 [1.0090, 7.4215]; held-out 384: 1.4507 [1.2023, 1.7424]	none	Tier 1
83	**not promoted**: lift_scan x Feistel order combined (this experiment only)	`koblitz-n83-combined-20261015`, run 38803625711	rT770, 576 new points	cold not measured	IC/rho 0.2145 [0.1949, 0.2353] (inverse of 4.6628 [4.2501, 5.1308], means); frozen primary R_m 1.2363 [1.0671, 1.4276], R_g 1.0611 [0.9083, 1.2430]: `not_separated`; combined vs B 1.9992 [1.7285, 2.3090]	none	Tier 1
83	promoted improvement over the combined producer: software-pipelined, prefetched root-table probes (this experiment only)	`koblitz-n83-probepipe-20261015`, run 38822490448	rT770, 384 new points	cold not measured	IC/rho 0.1462 [0.1307, 0.1626] (inverse of 6.8396 [6.1494, 7.6526]); combined producer 0.2047; online combined / pipeline 1.4001 [1.3971, 1.4032], probes identical	none	Tier 1
83	promoted improvement over the probe-pipeline producer: two-level root table, 32 MiB filter + 16-byte slots (this experiment only)	`koblitz-n83-tablefilter-20261015`, run 38829449594	rT770, 384 new points	cold not measured against rho; cold pP / pT 1.0101 [1.0090, 1.0113]	IC/rho 0.1523 [0.1358, 0.1709] (inverse of 6.5642 [5.8533, 7.3649]); pipeline 0.1620; online pP / pT 1.0633 [1.0589, 1.0679], probes identical, mostly from the 16-byte slot	none	Tier 1
83	promoted improvement over the combined producer: folded S3 root index, cold (this experiment only)	`koblitz-n83-state-fold-20261015`, run 38819494806	rT770, 16 new points	6.95 [5.53, 9.01] (cB 12.77; setup + online, rank stage excluded; with the rank stage 109.6, was 243.3)	IC/rho 0.163 [0.090, 0.241]; online 5% slower; cold cB / cF 1.8372 [1.8216, 1.8528], setup 1.86×, peak RSS 1.97×	none	Tier 1

Ratchet history

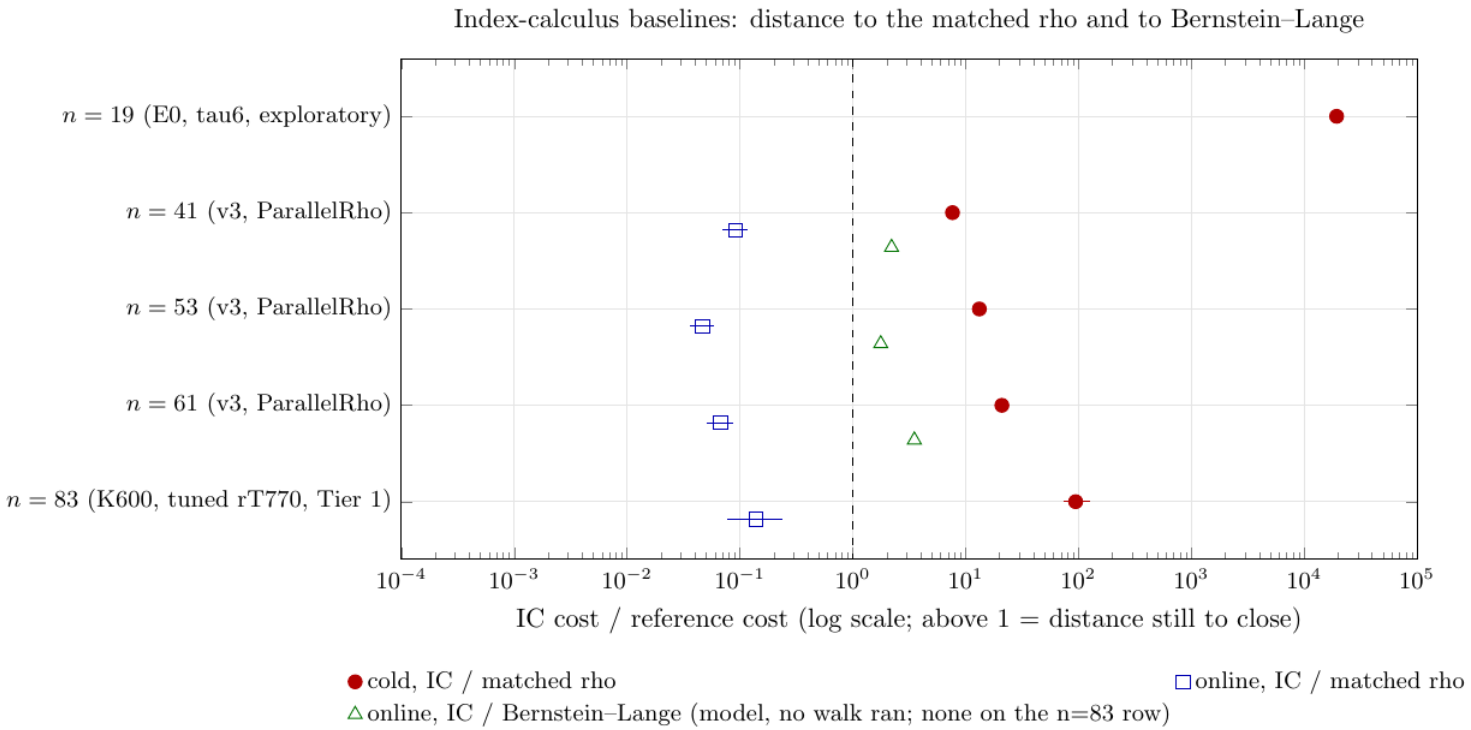


Figure 1: Distance to the matched rho and to Bernstein–Lange, per baseline row.

n	from	to	cold gap	online gap	class
41	§23 (v0, `0bf67f16`)	rule v3 (`995ea207`)	8.04 → 7.66	0.107 → 0.092	engineering (IC programme rounds R03, R05, R07)
53	§23 (v0)	rule v3	14.7 → 13.2	0.057 → 0.047	engineering
61	§23 (v0)	rule v3	31.6 → 21.0	0.113 → 0.067	engineering
83	B	lift_scan canonicalizer (PR #926, run 38776966036)	not measured	not measured (online 1.5152 faster, fresh 16)	engineering; promoted by its own frozen rule; baseline row not replaced (no cold `S`)
83	B	Feistel scan order (PR #927, run 38786433493)	not measured	not measured (online 1.4507 faster, held-out 384)	advance (probe count); promoted; baseline row not replaced (no cold `S`)
83	better single arm (lift_scan)	lift_scan x Feistel (run 38803625711)	not measured	R_m 1.2363 [1.0671, 1.4276]; R_g 1.0611 [0.9083, 1.2430]	**not promoted**: frozen primary `not_separated`
83	lift_scan x Feistel combined producer	software-pipelined, prefetched probes (run 38822490448)	not measured	0.2047 → 0.1462 (online, tuned rho; 1.4001 faster)	engineering; promoted by its own frozen rule; baseline row not replaced (no cold `S`)
83	probe-pipeline producer	two-level root table (run 38829449594)	not measured	0.1620 → 0.1523 (online, tuned rho; 1.0633 faster)	engineering; promoted by its own frozen rule; baseline row not replaced (no cold `S` against rho)
83	lift_scan x Feistel combined producer	folded S3 root index (run 38819494806)	12.77 → 6.95 (rank stage excluded)	not improved (online 5% slower)	engineering; promoted by its own frozen rule; baseline row not replaced (rank stage still dominates cold cost)
83	B (BL-arm baseline row, run 38689817618)	stacked producer: probe pipeline + 16-byte slots + folded index + rank stage (run 38862448935)	360 → 94.4 (end-to-end cold; different run and instances)	1.01 → 0.139	engineering; promoted by its own frozen primary (end-to-end cold cP / cS 3.2391 [2.9809, 3.5240], 20 points, two rank blocks, against the probe-pipeline producer); **baseline row replaced**: the first measured n=83 end-to-end cold `S` after B (9.68; was 32.93 in B's run, a different step time)

The paired cold-time ratios between baselines, each in its own runs, are in crypto `research/notes/index-calculus/IC_TOOL_PROGRAM.md` §7. The gaps above are each against that comparison's own rho, and that rho also changed between §23 and v3. So the gap delta is the distance remaining, not the IC's own speedup.

Stage results: canonicalization

Stage improvements are recorded here. One becomes a pipeline improvement only once it moves an end-to-end `S` above.

stage	n	experiment	result	tier	pipeline effect
normal-basis least rotation, isolated	131	`canon-n131-20261014`	Five exact canonicalizers on `[u64; 3]`. Ir per uniform word: lift_scan 514, gap_scan 1,435, exhaustive 1,850, exhaustive_bf 3,664, refine_bf 4,510. 0 mismatches.	deterministic Ir; wall times exploratory	none yet
the same, inside a decomposition-style loop	131	`canon-n131-loop-20261014`, run 38607684318	lift_scan is fastest on all four workloads: 16/16 contrasts at Bonferroni, ratios 1.83–3.81. In-loop Ir over the loop floor: lift_scan 519, gap_scan 1,760, exhaustive 2,883.	Tier 1 for the timing	none yet. No n=131 pipeline exists, and at n=53 the isolated ranking did not survive the loop (`papers/frobenius-orbits` §4/§7)

Open

- **n=41, 53, 61:** the gap to the tuned rho is measured by `experiments/koblitz-s23-tuned-rho-20261015` (run 38777903922, PR #923; rows above, one per size, each from that experiment alone; n=47, 57 and 59 are in its README). It ran rT770's configuration on the one-word `Gf2` kernel, because `Gf2_128` exists only for $n > 64$, and it measures about the same cost as ParallelRho. It is accounting: no IC change, so no baseline row moves. A rho tuned on the one-word kernel is not measured.
- **n=19 and n=41–61:** no Bernstein–Lange walk has run, so the BL figures are models or absent.
- **n=83:** the baseline row is the stacked producer (`koblitz-n83-stacked-20261016`, run 38862448935): the probe pipeline, 16-byte slots, folded index and a rank stage on the stack's own probe loop, promoted end to end over the probe-pipeline producer (3.2391 [2.9809, 3.5240], 20 points, two rank blocks). Its end-to-end cold `S` is 9.68 against the tuned rho's 0.103 (gap 94.4 [73.5, 125.3]); it is the first n=83 row with a measured end-to-end cold `S` after B. B's earlier row (experiment on PR #918, not merged yet) moved to Other verified records. The parts were each promoted on their own parent: lift_scan and the Feistel order over B (their combination is not promoted over the better single arm, `koblitz-n83-combined-20261015`), the probe pipeline over the combined producer (`koblitz-n83-probepipe-20261015`), the two-level root table over the probe pipeline (`koblitz-n83-tablefilter-20261015`, 1.0633, mostly the 16-byte slot) and the folded index over the combined producer (`koblitz-n83-state-fold-20261015`, cold 1.8372). End to end, neither the fold nor the slots is separated from 1 inside the stack. The rank stage is 91% of the end-to-end cold cost, and the stack's online is about 4.6% slower than the probe pipeline's. The BL gap was not re-measured on the stack.
- **n=19:** there is no IC1 candidate manifest, and online `S` is not separable.

Updating this file

1. Freeze a protocol for the candidate improvement against the row's pipeline, on the same instances, before timing.
2. When the improvement lands, replace the row in `baselines.json` and in the table above. Add a history row with the before and after figures and the class (advance, engineering, relabelling, accounting). Relabelling and accounting do not replace a baseline.
3. Name the next improvement to try in the new row.
4. Re-render Figure 1 and this file's PDF:

```
cd docs/ic/figures && pdflatex baselines_gap.tex && pdftoppm -png -r 160 -singlefile baselines_gap.pdf baselines_gap
cd .. && pandoc BASELINES.md -o BASELINES.pdf --pdf-engine=pdflatex -H pdf-header.tex -V geometry:landscape,margin=1.2cm -V fontsize=9pt
python3 -m json.tool baselines.json > /dev/null
```

Sources

- crypto `docs/ic/BOUNDARY_TARGETS.md` and `docs/ic/boundary_targets.json` (Regime B `vs_rho`; complete cold compact-orbit panel).
- crypto `research/notes/index-calculus/RESEARCH_IC_BOUNDARY_LEDGER.md` §23; `research/ic_single_target_20260930/` (PROTOCOL, analysis.json).
- crypto `research/ic_tool_program/rule/v3/` (README, PROTOCOL, analysis.json); `research/notes/index-calculus/IC_TOOL_PROGRAM.md` §7.
- crypto `research/notes/ecc2k130/disjoint_cold_v2_outcome_20261001/RESULT.md`.
- cryptanalysis `experiments/koblitz-n83-bl-arm-20261014/` (branch `claude/n83-bl-arm`, PR #918): README, `run_github_38689817618/result.json`.
- cryptanalysis `experiments/koblitz-s23-tuned-rho-20261015/`: README, `run_github_38777903922/panel/result.json` (PR #923).
- cryptanalysis `experiments/koblitz-n83-reconcile-20261014/`: README, `run_github_38617911080/panel/result.json`.
- cryptanalysis `experiments/koblitz-single-target-n83-tier1-20261014/`: README, `tier1_github_38523707244/`.
- cryptanalysis `experiments/volcano-ic/rho-arm-20261014/`: README, `results/summary.json`.
- cryptanalysis `experiments/canon-n131-20261014/` and `experiments/canon-n131-loop-20261014/`.